

Aaron Lin

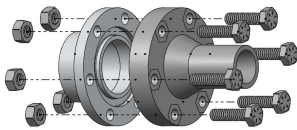
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Rocket hardware, autonomous prototypes, and testing

B.S. Mechanical Engineering, Aerospace Concentration · Boston University · May 2028

I am a mechanical engineering junior seeking a **Summer 2027 internship** in mechanical design, aerospace, or product development. I have sized and fabricated rocket hardware, written controls for an autonomous robot, and compared laboratory measurements with analytical models.

Selected engineering work

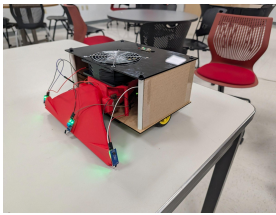


Converging-diverging rocket nozzle

My role: individual mechanical lead · Completed
Flow sizing, CAD, drawings, pressure analysis, resin
printing, and static-fire integration.

2.12

minimum factor
of safety in the
pressure-only model



The Deskinator

My role: firmware and system integration · Team of four
Arduino navigation, edge sensing, gesture activation,
troubleshooting, and iterative pickup testing.

76%

pickup rate in the
documented final trial



ARGO aerodynamic fairing

My role: project engineer · In progress
Installed-geometry fairing concepts for an external
downcomer and wiring at a Mach 2.81 design condition.

CDR complete

September 2026;
comparative analysis
in progress

Experience and capabilities

Mechanical Engineering Intern, Culturon

Sydney, Australia · February–June 2026

Ran optical emission spectroscopy experiments and built Python tools with pandas, SciPy, and Astroquery to process plasma spectra and compare candidate lines with the NIST atomic spectra database.

Project Engineer, BU Rocket Propulsion Group

September 2024–present

Designed and fabricated a resin rocket nozzle; now comparing aerodynamic fairing concepts using ANSYS Fluent, OpenRocket, and first-principles calculations.

Design and analysis

SolidWorks · ANSYS Fluent · OpenRocket · FEA ·
CFD · GD&T fundamentals

Programming and data

Python · pandas · SciPy · Astroquery · MATLAB ·
Arduino C++

Fabrication and test

Resin and FDM printing · CNC and manual
machining · soldering · sensors · experimental testing

More drawings, reports, test media, and complete case studies: aaronlin.space.

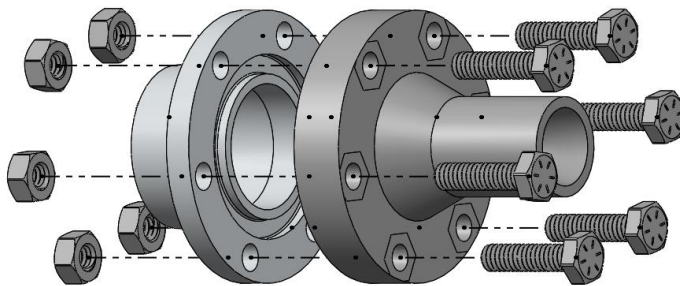
Converging-diverging rocket nozzle

BU Rocket Propulsion Group · Fall 2024 · Completed

My role: Individual mechanical lead **My contribution:** Flow sizing, CAD, drawings, pressure analysis, fabrication, and integration

Team outcome: Integrated with the chamber and ignition system for static-fire testing **Evidence status:** Pressure model and build artifacts documented; performance not measured

I owned the nozzle's mechanical design and fabrication. I sized the flow path with one-dimensional compressible-flow relations, produced the CAD and drawings, printed the parts, and integrated the assembly for static-fire testing. The design used an 80 psi chamber-pressure requirement, 3250 K chamber temperature, 13 g/s mass flow, and a 2.1 psia target exit pressure.



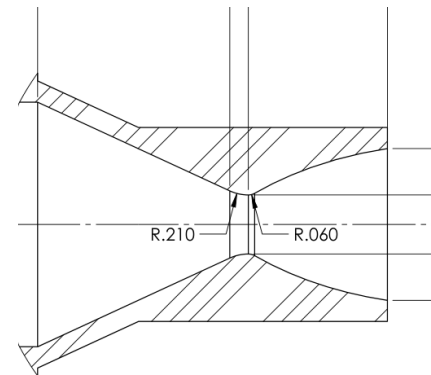
Nozzle, six-bolt adapter, and fasteners.

6.521

exit-to-throat area ratio

2.12

minimum factor of safety; pressure-only model



Internal flow path and controlling dimensions.

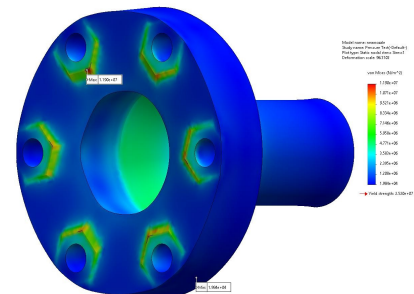
My decisions

I used isentropic-flow relations to size the throat and exit, kept the assembly within a 1.05 in diameter limit, and modeled the nozzle and adapter in SolidWorks. I selected resin printing for the internal geometry and schedule, produced the drawings, fabricated the nozzle, integrated it with the chamber and seals, and helped assemble the relay ignition circuit.

Pressure-analysis evidence

The SolidWorks Simulation model applied approximately 46.3 psi at the throat and 80 psi in the chamber. It reported a maximum von Mises stress of 11.90 MPa against an assumed 25.20 MPa yield strength.

Limits: The 2.12 factor of safety applies only to the pressure model and its material and boundary assumptions. The hot-fire recording demonstrates system integration, not quantified nozzle performance.



Pressure-only contour: 11.90 MPa maximum stress versus 25.20 MPa assumed yield; deformation exaggerated.

Next validation. A second iteration should add temperature-dependent properties, heat transfer, mesh convergence, pressure and wall-temperature measurements, and a documented post-fire inspection.

Methods: SolidWorks · FEA · one-dimensional compressible flow · resin printing

[Open the full nozzle case study](#)

The Deskinator

Boston University EK210 · Team of four · Fall 2025 · Completed

My role: Firmware and system integration **My contribution:** Arduino navigation, sensing, activation, assembly, and troubleshooting

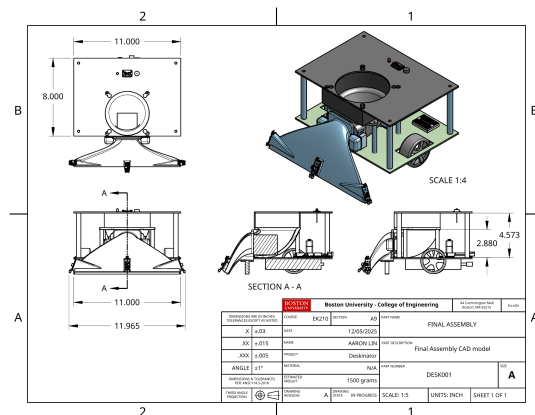
Team outcome: Pickup improved after fan, filter, and nozzle changes **Evidence status:** Pickup quantified; edge response demonstrated but not fully repeatable

Our team built an autonomous robot intended to clean a 4 ft by 4 ft desk without driving over an edge. I wrote the Arduino C++ firmware for S-curve pathing, IR-triggered edge avoidance, and gesture activation, and helped assemble and troubleshoot the mechanical and electrical systems.

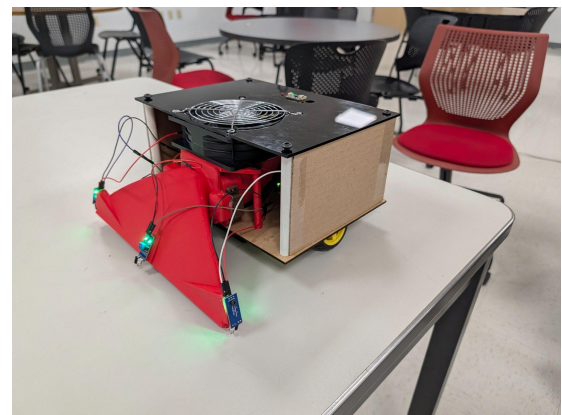
76%

pickup rate in the documented final trial

The course report records a top-four class demonstration finish among roughly 20 groups.



Final assembly drawing with fan, filter, battery, electronics, and nozzle.



Prototype used for presentation and pickup testing.

Iteration driven by testing

Drive. Motors struggled under the robot's weight; the team replaced them with 12 V N20 encoder motors, restoring controlled movement.

Vacuum. Early tests showed the original fan could not collect rice reliably; the team installed a stronger fan.

Filter. A delayed replacement filter threatened testing; I helped build a perforated paper-cup-and-gauze substitute.

Nozzle. Limited debris contact reduced pickup; the opening was narrowed and angled toward the desk.

My firmware and integration work

The firmware waits for a recognized gesture, enables the fan and motor driver, and runs a timed routine. It reads three downward-facing IR sensors before each command, stops both motors at an edge, and uses encoder counts to control the pivot.

Limits: The report records an early trial near 24% pickup and a final trial with 72 of 95 grains collected after the fan, filter, and nozzle changes. Edge avoidance and gesture activation were demonstrated but not fully repeatable. The next test should repeat timed pickup trials across several debris patterns and record both attempted and successful edge detections.

Methods: Arduino C++ · IR and gesture sensors · CAD · prototype testing

[Open the full Deskinator case study](#)

ARGO aerodynamic fairing

BU Rocket Propulsion Group · Project Engineer · July 2026–present · In progress

My role: Project engineer **My contribution:** Baseline and candidate geometry, stagnation calculations, material screening, and CFD preparation

Team outcome: Conceptual Design Review completed **Evidence status:** Current milestone documented; comparative simulation results pending



Installed packaging model showing ARGO's external downcomer and an early aft-fairing concept.

Design problem

ARGO routes a downcomer and wiring along the rocket exterior, creating aerodynamic, thermal, attachment, and packaging challenges. The fairing must protect those components while limiting drag and remaining compatible with the installed geometry.

My work

I am comparing the previous Icarus aerocover with candidate leading edges, midsections, and trailing-edge or boattail concepts using ANSYS Fluent, OpenRocket, and first-principles calculations. I translate the installed packaging constraints into consistent baseline and candidate geometries and document the assumptions used in each comparison.

Completed milestone

The team completed Conceptual Design Review on September 3, 2026. At the **Mach 2.81 design condition**, work to date includes stagnation temperature and pressure calculations, early material screening, and preparation of candidate CFD geometry.

Comparison controls

A fairing attached to the rocket body is only a partial revolution, so results from a fully revolved nose cone do not transfer directly. Baseline and candidate concepts must use the same installed geometry, boundary conditions, reference area, and reporting method before their aerodynamic results can be compared.

Limits: This is an in-progress design review, not a completed performance claim. No drag reduction, pressure result, or selected geometry is reported until the baseline-versus-candidate simulations are complete and checked for mesh and setup consistency.

Next analysis

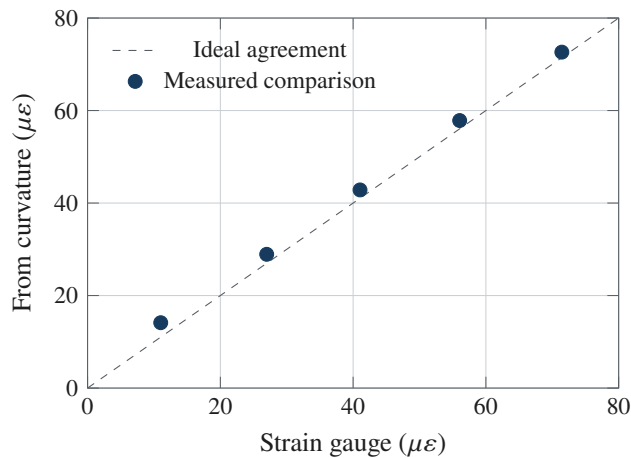
Before Preliminary Design Review, I will run the baseline and candidate with matched geometry, boundary conditions, reference area, and reporting methods. I will document the mesh and compare one checked performance metric.

Methods: ANSYS Fluent · OpenRocket · supersonic flow · first-principles calculations

[Open the current ARGO case study](#)

Experimental and electronics work

Structural mechanics, instrumentation, analysis, and CAD



Increasing-load measurements compared with the ideal one-to-one relationship.

Structural mechanics laboratory studies

University of Sydney · Spring 2026 · Shared laboratory measurements

At the largest applied load, the strain gauge measured **71.44 microstrain** and the curvature calculation predicted **72.63 microstrain**, a **1.63%** difference. At 222.41 N, measured deflection reached 1.38 mm at one hook and 1.99 mm near midspan; the report's beam model predicted the measured values within about 10% across the load steps.

Authorship: I used measurements gathered with classmates in my calculations and reports; the measurements are presented as team evidence.

[Open the structural mechanics case study](#)



Room temperature monitor

Boston University EK131 · Spring 2025 · Individual build

I built and programmed an Arduino Uno monitor with a TMP36 sensor, 16 by 2 LCD, LEDs, and a buzzer. It displays Celsius and Fahrenheit and signals readings outside 60–75 °F. The circuit drew 76 mA; a 690 mAh battery gives an ideal estimate near nine hours.

Limits: The sensor was not calibrated against a reference thermometer, and full runtime was not measured. The build demonstrated sensing, dual-unit display, and high- and low-temperature alerts.

[Open the temperature-monitor case study](#)

Additional work online

[Diffusion analysis](#)

Sample A produced a 23.6 kJ/mol activation-energy fit; sample B remained inconclusive.

[Acrylic truss](#)

The final MATLAB model predicted 46.59 oz; the physical build met the 16 oz minimum.

[Chiller analysis](#)

Calculated COP was 5.437 at a 5 °F approach versus 3.6109 at 15 °F.

[SolidWorks exercises](#)

Parts, assemblies, section views, exploded views, drawings, bills of materials, and motion studies.

See the full case studies, drawings, reports, and test media at aaronlin.space.

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